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System and Method for Modeling Stress Disturbances for Multi-Lateral Wellbore Drilling

Osman Hamid and Hussain Albahrani, Saudi Aramco; Eslam Abdalrahman, King Fahd University of Petroleum and Minerals; Rowa Tawfiq, Saudi Aramco

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Abstract

This study introduces an innovative system and methodology for modeling stress disturbances during the drilling of lateral wellbores in subsurface formations, addressing challenges associated with multilateral wellbore construction. The primary objectives are to accurately predict stress-induced deformation, evaluate potential failure risks, and optimize drilling parameters to ensure wellbore stability and operational efficiency.

The methodology employs a sequential finite element analysis (FEA) approach to simulate the formation of a main wellbore and subsequent lateral branches. The system integrates a stress field assessment engine, incorporating pre-processing components for assigning geomechanical properties, a simulator for stress field calculations, and a model updating module to include lateral wellbore geometries. The stress field simulations account for in-situ stresses, material properties, and disturbed stress fields, while employing advanced failure criteria, such as Drucker-Prager and Mohr-Coulomb, to evaluate deformation risks. The use of 3D meshing with 20-node brick elements ensures high-resolution calculations and computational efficiency.

Results demonstrate that the proposed methodology successfully predicts stress distributions and failure risks in real-time drilling scenarios. The simulations identify optimal mud weight and pressure parameters, reducing the likelihood of shear or tensile failures and ensuring wellbore integrity. The methodology also highlights the critical influence of stress disturbances from the main wellbore on lateral wellbore stability.

The conclusion emphasizes that this system provides a robust framework for modeling and mitigating stress disturbances in multilateral wellbores. By accurately simulating real-world drilling conditions and offering actionable insights into stress-induced risks, the system enhances operational efficiency and reduces costs.

The innovation lies in the sequential stress field modeling approach, which retains and builds upon initial stress field data from the main wellbore to predict cumulative effects during lateral wellbore formation. This step-change in geomechanical modeling capabilities offers a transformative tool for optimizing multilateral wellbore design and

Methodology

Directional drilling, commonly referred to as slant drilling, involves the creation of non-vertical boreholes to optimize subsurface access to hydrocarbon-bearing formations. A significant advancement in this practice is the deployment of lateral wells, which extend from a primary vertical or deviated borehole known as the main or mother wellbore into the formation. When more than one such lateral is drilled from a single main borehole, the resulting configuration is termed a multilateral well. This approach enhances reservoir contact, improves production rates, and increases hydrocarbon recovery efficiency.

In conventional operations, it is often assumed that lateral branches drilled from a common main wellbore will exhibit similar drilling conditions, especially when they are located in the same formation, follow similar trajectories, and employ identical drilling tools. However, field data and operational feedback challenge this assumption. Drillers frequently encounter unexpected stress-related complications, particularly during the initiation of lateral branches. These issues arise due to stress field disturbances caused by the drilling of the main wellbore itself. The pre-drilled mother wellbore alters the in-situ stress conditions of the surrounding rock, creating a perturbed stress environment that directly influences the mechanical response of the formation during subsequent lateral drilling. [Zoback, 2010](#); [Jaeger et al., 2007](#) evaluated potential failure risks, and optimize drilling parameters to ensure wellbore stability and operational efficiency, but their studies focus on a single wellbore. The pre-drilled mother wellbore alters the in-situ stress conditions of the surrounding rock, creating a perturbed stress environment that directly influences the mechanical response of the formation during subsequent lateral drilling ([Chen and Tan 2006](#), [Plumb, 1991](#); [Schmitt et al., 2012](#) and [Zhang 2013](#)).

To address this gap, a computational framework is proposed in this study that accurately captures the sequential nature of drilling operations and the resulting evolution of stress fields. The system implements a finite element analysis (FEA)-based modeling engine that initially simulates the drilling of the main wellbore and calculates its induced stress perturbations. The model is then updated to introduce lateral wellbores, allowing for a dynamic and realistic simulation of stress evolution.

This framework offers several key advantages:

- It more closely mimics real drilling sequences by retaining the stress redistribution caused by the mother wellbore.
- It allows for stress-based failure analysis using various mechanical failure criteria such as the Mohr-Coulomb, Drucker-Prager, and Mogi-Coulomb models ([Aadnoy, 1990 and 2006](#) and [Mogi, 1971](#)).
- It enables optimized mud weight recommendations for both main and lateral wellbores to prevent tensile or shear failures ([Morita et al., 1990](#))

Methodology

The methodology employs a sequential finite element analysis (FEA) approach to simulate the formation of a main wellbore and subsequent lateral branches. The system integrates a stress field assessment engine, incorporating pre-processing components for assigning geomechanical properties, a simulator for stress field calculations, and a model updating module to include lateral wellbore geometries. The stress field simulations account for in-situ stresses, material properties, and disturbed stress fields, while employing advanced failure criteria, such as Drucker-Prager ([Drucker & Prager, 1952](#)) and Mohr-Coulomb ([Aadnoy, 2006](#)), to evaluate deformation risks. The use of 3D meshing with 20-node brick elements ensures high-resolution calculations and computational efficiency.

Results demonstrate that the proposed methodology successfully predicts stress distributions and failure risks in real-time drilling scenarios. The simulations identify optimal mud weight and pressure parameters, reducing the likelihood of shear or tensile failures and ensuring wellbore integrity ([Zhang, 2013](#)). The

methodology also highlights the critical influence of stress disturbances from the main wellbore on lateral wellbore stability (Zhang et al., 2005).

Initial Model Generation Pre-processing Stage

This phase sets up the simulation domain, stress boundary conditions, and material behavior before any actual solving of equations takes place. It prepares the model to simulate wellbore integrity, stress redistribution, and failure onset.

Fig. 1 presents a detailed and structured methodology for modeling stress redistribution and optimizing drilling mud weight in multilateral wellbore systems. The approach integrates Finite Element Method (FEM) simulations with geomechanical failure criteria to ensure safe and effective wellbore design and operation. The simulation begins with a comprehensive definition of input parameters, including:

- **Geomechanical Properties:** Young's modulus, Poisson's ratio, Unconfined Compressive Strength (UCS), friction angle, and dilation angle.
- **Geometry:** Trajectories of the main and lateral wellbores, junction depths, and borehole curvature.
- **In-situ stresses:** Overburden stress (σ_v), maximum horizontal stress (σ_{Hmax}), minimum horizontal stress (σ_{Hmin}), and formation pressure (P_0)
- **FEM Mesh Generation** A 3D domain encompassing the wellbores and surrounding rock is discretized using 20-node isoperimetric hexahedral (brick) elements:
 - Ensures higher-order interpolation for curved geometries.
 - Provides high spatial resolution of stress concentration near wellbores.
 - Uses adaptive meshing, with finer elements around the borehole and junction.
- **Simulation of Initial Model:** The initial model simulates the formation of the main wellbore:
 - FEM solves for stress distribution using governing equations given by Hooke's law:

$$\sigma_{ij} = D_{ijkl} \varepsilon_{kl} \quad (1)$$

Where:

σ_{ij} is the second -order stress tensor

ε_{kl} is the second -order strain tensor

D_{ijkl} is the fourth-order elasticity tensor

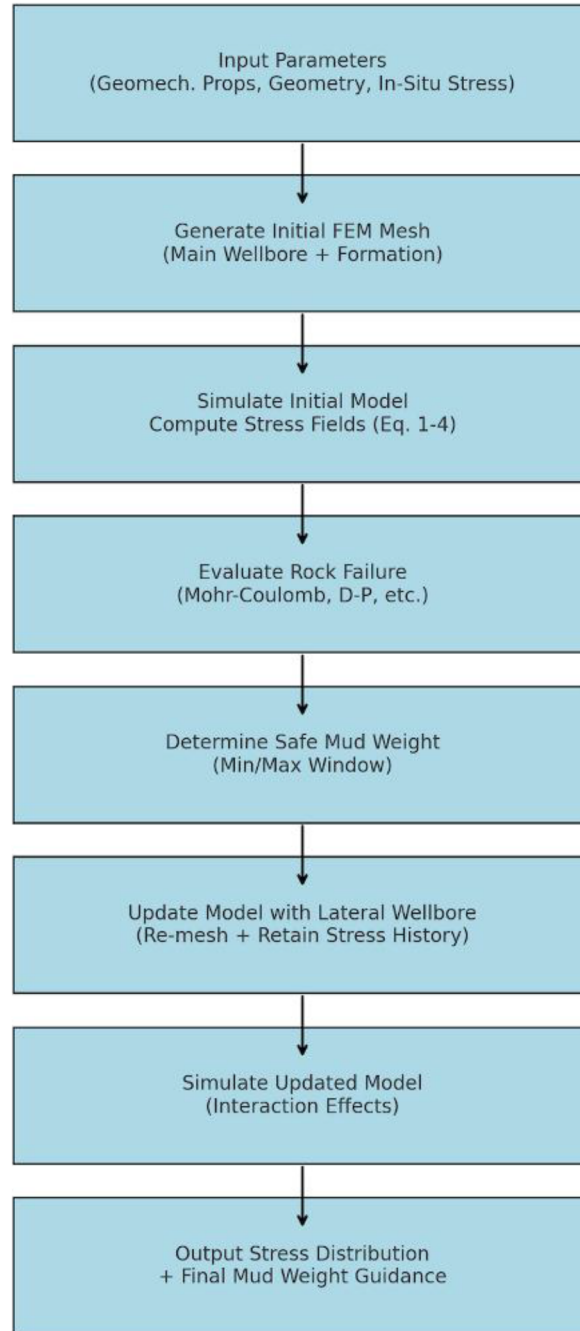


Figure 1—FEM-Based Stress Modeling Workflow for Multilateral Wellbores (Hamid et al. 2024)

Displacement field can be expressed by the weak form of equilibrium:

$$\int_{V^e} B^T D B d\Omega = \int_{V^e} N^T F d\Omega - \int_{S^e} N^T T d \quad (2)$$

Where:

B: Strain-displacement matrix (relates displacements to strains)

D: Constitutive (material stiffness) matrix

N: Shape function matrix

F: Body force vector (e.g., gravity)

T: Traction vector on boundaries

V^e : Volume of an element

S_e : Surface of an element

- Plasticity is handled via associated plastic flow rules:

$$\Delta \varepsilon_{ij}^p = \lambda \frac{\partial \phi(\sigma_{ij})}{\partial \sigma_{ij}} \quad (3)$$

Where:

$\Delta \varepsilon_{ij}^p$ is the increment in plastic strain

λ : Plastic multiplier (non-negative, governs magnitude of plastic flow)

$\phi(\sigma_{ij})$: Plastic potential function (e.g., same as yield function in associative flow)

- **Failure Evaluation:** Post-processing evaluates stress state against rock failure criteria:
 - Material behavior is modeled using Drucker-Prager or Mohr-Coulomb criteria.

$$\phi(\sigma_{ij}) = \sqrt{\frac{1}{2} S_{ij} S_{ij}} - a_0 + a_1 \sigma_m = 0 \quad (4)$$

- Mohr-Coulomb: For frictional failure.
- Failure envelope is checked at each node.
- Zones at risk of tensile or shear failure are flagged.
- **Mud Weight Window Determination:** Based on failure analysis:
 - Minimum mud weight

$$MW_{min} = \frac{\sigma_\theta - S}{0.052D} \quad (5)$$

Maximum mud weight

$$MW_{max} = \frac{\sigma_r + T}{0.052D} \quad (6)$$

Where:

σ_θ and σ_r : Tangential (hoop) and radial stress around the borehole

S and T : Shear and tensile strength or failure threshold (e.g., cohesion or Mohr-Coulomb limit)

D : True vertical depth (TVD) in feet

0.052: Unit conversion factor to relate pressure gradient (psi/ft) to mud weight (ppg)

This defines the operational mud window to ensure wellbore stability

- **Model Update with Laterals:** Lateral wellbores are added to the model:
 - The stress field from the main borehole is retained.
 - Geometry is re-meshed at junctions.
 - New FEM mesh accounts for spatial and angular distribution of laterals.
- **Updated Model Simulation:** The updated model captures:
 - Interaction of stress pulses and wavefronts.
 - Concentration zones near junctions.
 - Redistribution of effective stress around laterals.

- Additional failure and deformation zones.
- Output and Final Mud Guidance: Final results include:
 - 3D distribution of radial and tangential stresses.
 - Failure risk maps.
 - Updated safe mud weight envelope.
 - Exported datasets for operational planning and FEM validation.
- Wave propagation and stress pulse superposition Modified after (Carcione 2007):

Radially decaying stress pulse:

$$\sigma_{pulse}(r, t) = \frac{A}{r} e^{-\alpha r} \sin(\omega t - kr) \quad (7)$$

A: Source amplitude

α : Spatial attenuation coefficient

ω : Angular frequency

k : Wave number

r : Radial distance from source

Multilateral Geometry Stress Superposition

For multilateral wellbores, each lateral induces a stress pulse. The stress field is a superposition of pulses from each branch. Let's assume n laterals radiating from the mother wellbore, each contributing a stress disturbance $\Delta\sigma_i$:

$$\sigma_{r,total}(r, \theta, t) = \sum_{i=1}^n \sigma_{pulse,i}(r_i, \theta_i, t) + \Delta\sigma_r^{(i)}(r, \theta - \theta_i) \quad (8)$$

$$\sigma_{\theta,total}(r, \theta) = \sum_{i=1}^n \sigma_{pulse,i}(r_i, \theta_i, t) + \Delta\sigma_{\theta}^{(i)}(r, \theta - \theta_i) \quad (9)$$

Where each $\Delta\sigma^{(i)}$ can be approximated using modified Kirsch-type or acoustic-based pulses.

The attenuated stress amplitude $\sigma(r, t)$ can be modeled as

$$\sigma(r, t) = \sigma_0 \cdot e^{-\alpha r} \cdot e^{-\beta t} \quad (10)$$

Stress Interference Formulation

The total stress at a point (r, t) due to multiple laterals emitting transient stress pulses is computed by superposing individual stress pulse contributions:

$$\sigma_{total}(r, t) = \sigma_0 + \sum_{i=1}^n A_0^{(i)} \cdot e^{-\alpha|r-r_i|} \cdot e^{-\beta t} \quad (11)$$

Combining equations 8 to 11 and adding dynamic boundary conditions, the unified stress pulse interference formulation that synthesizes radial and tangential stress interference due to multiple stress pulses emitted from laterals in a formation can be given by:

$$\sigma_{\alpha,total}(r, \theta, t) = \sigma_0 + \sum_{i=1}^n A_0^{(i)} \cdot e^{-\alpha|r-r_i|} \cdot e^{-\beta t} \cdot f_{\alpha}(\theta - \theta_i) \cdot g_{\alpha}(\theta) \quad (12)$$

Where:

$\sigma_{\alpha,total} \in \{\sigma_r, \sigma_{\theta}\}$: Total radial or tangential stress component.

σ_0 : Far-field stress (psi) for the first lateral L_1 .

$A_0^{(i)}$: Initial pulse amplitude at lateral i (psi).

r_i : Radial distance of lateral i (m).

θ, θ_i : Observation and lateral azimuth (radians or degrees)

α : spatial attenuation factor (1/m).

β : temporal attenuation factor (1/s or 1/hr).

t : time since pulse emission (s or hr).

$f_a(\theta - \theta_i)$: Directional influence function (e.g., $\cos(\theta - \theta_i)$, or anisotropic kernel).

For radial stress:

$$f_r(\theta - \theta_i) = \cos^2(\theta - \theta_i) \quad (13)$$

For tangential stress:

$$f_\theta(\theta - \theta_i) = \sin^2(\theta - \theta_i) \quad (14)$$

Directional dynamic boundary conditions:

$$g_i(\theta) = \begin{cases} \frac{A_{2i}^{-1}}{A_1} & \text{for } \theta < \theta_i \text{ (Interfering side)} \\ \frac{A_{2i}}{A_1} & \text{for } \theta \geq \theta_i \text{ (dissipative side)} \end{cases} \quad (15)$$

The Finite Element Method (FEM) is implemented using a modular structure consisting of 33 subroutines, rewritten by a central driver code. Among these, twelve (12) main subroutines are directly invoked by the driver to execute key computational tasks (Hamid et al. 2024). These include reading the input files, applying mechanical loads, constructing and assembling the global stiffness matrix, and solving the resulting system of equations. The interaction between the driver and its associated subroutines is illustrated in the flowchart presented in Fig. 2. Once the system of equations is solved following the formulation in Equation (1) the displacement vector u is obtained. This enables the computation of residual forces, which are used to assess both equilibrium and numerical convergence. This is done by subtracting the right-hand side vector (body and traction forces) from the left-hand side, which is the product of the global stiffness matrix K and the displacement vector u . Ideally, this residual should be zero to satisfy the equilibrium condition exactly. However, due to numerical limitations, a small tolerance value is introduced to define an acceptable convergence criterion.

If the residual norm falls below the specified tolerance threshold commonly set close to but not exactly zero convergence is considered achieved. Otherwise, the residual is carried forward into the next iteration. This iterative procedure is embedded within a nested loop framework: the convergence loop operates inside the load increment loop, both of which are delineated in Fig. 2. Load increments are predefined in the input file and govern the progression of the solution through successive equilibrium states.

In the disciplines of rock mechanics and geotechnical engineering, numerous validated constitutive models have been established to delineate failure criteria for both rock and soil media. These models are predominantly formulated based on empirical data obtained from laboratory experiments and are designed to characterize the failure envelope a boundary surface in stress space beyond which material failure is anticipated.

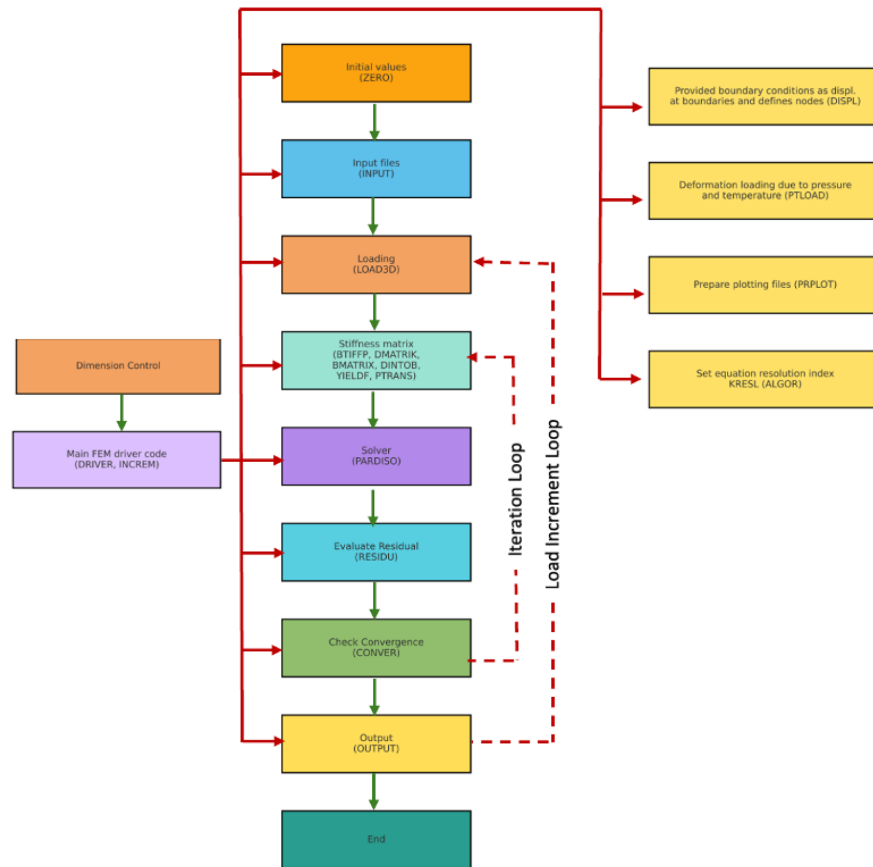


Figure 2—Flow chart of the driver code of the FEM and the 12 main subroutines (Hamid et al. 2024).

The failure envelope is defined in terms of material-specific strength parameters, such as cohesion, internal friction angle, or critical stress invariants, which vary depending on the adopted failure model. These strength parameters govern the permissible combinations of shear and normal stresses that a rock or soil specimen can withstand under triaxial loading conditions. Failure is predicted to occur when the computed in-situ stress state exceeds this envelope, indicating that the material has reached or surpassed its compressive strength limit.

Among the most widely implemented failure models in geomechanical simulations are the Mohr-Coulomb, Mogi, Drucker-Prager, and Lade criteria. These models serve as quantitative tools for assessing whether a given stress state exceeds the failure threshold defined by material-specific strength parameters. In finite element analysis (FEM), this evaluation is conducted at each discretized point (node or element) by comparing the in-situ stress state with the calibrated failure envelope.

For instance, when applying the Mogi criterion, the three principal stresses ($\sigma_1, \sigma_2, \sigma_3$) derived from FEM are used to compute the octahedral shear stress (τ_{oct}), which characterizes the deviatoric stress state. A scalar failure function f , dependent on the local strength parameters, is then evaluated. The stability assessment is performed by subtracting f from the computed (τ_{oct}). If the result is positive, this indicates that the local stress state lies outside the failure envelope, and thus, failure is anticipated at that location.

This procedure is summarized in the decision flowchart depicted in Fig. 3, which visualizes the evaluation logic for each conventional failure model. It is important to emphasize that throughout this formulation, compressive stresses are conventionally treated as negative values, ensuring consistency with geomechanical sign conventions.

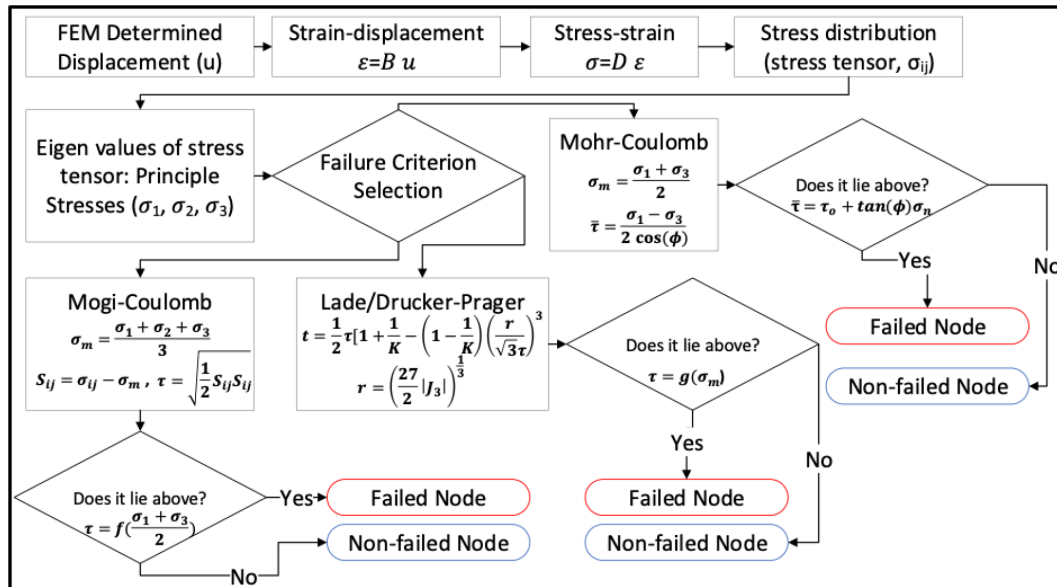


Figure 3—A flow diagram describing the process for implementing conventional failure criteria into finite element modelling (Hamid et al. 2024).

Results

Fig. 4 illustrates a 3D Finite Element Method (FEM) multilateral wellbore model, incorporating the geometric layout of a main vertical well (L0) and four lateral branches (L1 through L4) spaced at 25° intervals from a junction point. The figure shows stress Interference and Propagation Concept, the translucent red and yellow spheres around L0 and L1 represent zones of stress perturbation or stress pulses due to wellbore excavation.

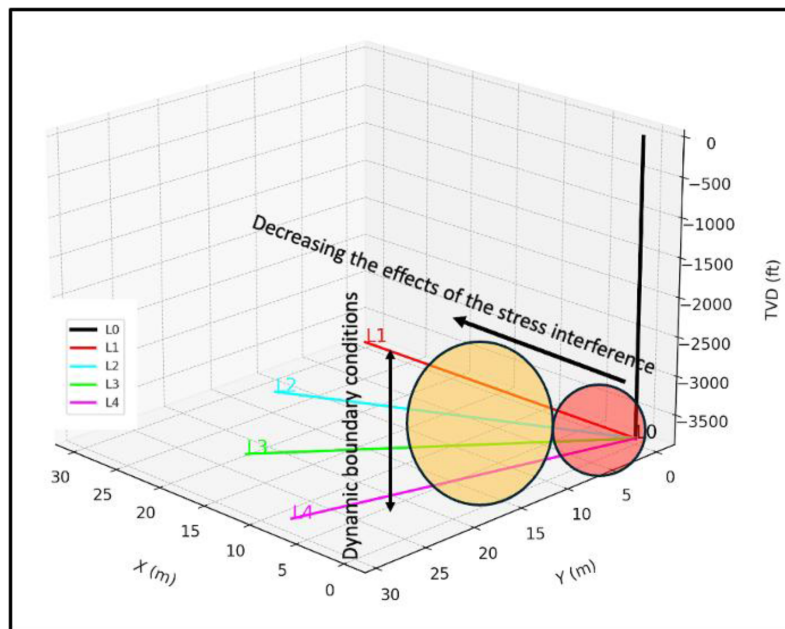


Figure 4—3D FEM flow: Multilateral wellbore geometry (Modified after Hamid et al. 2024)

The arrow labeled "Decreasing the effects of the stress in"erference" points away from the motherbore (L0), suggesting that, the magnitude and overlap of stress fields diminish with increasing radial distance

from the junction. Proper spacing and angular distribution of laterals help mitigate mechanical interference and stress shadowing among laterals.

The figure also shows dynamic Boundary Conditions, where the vertical arrow labeled "Dynamic boundary conditions" emphasizes that boundary constraints (displacement or force) evolve as additional laterals are drilled. These conditions govern stress redistribution in the formation and are crucial in FEM-based geomechanical models. These model approaches enhance analyzing wellbore stability under complex multi-lateral geometries, stress field superposition and failure risk zones using criteria like Mohr-Coulomb or Drucker-Prager.

Fig. 5 illustrate tangential and radial stress distributions accounting for dynamic boundary conditions and asymmetric energy dissipation from multilateral wellbores using equation 12. The stress distribution mainly depends on stress anisotropy, wellbore azimuth with respect to the horizontal stress orientation, and stress regime. L1 lateral assume to be drilled in the direction of the maximum horizontal stress in strike-slip fault regime, the stress anisotropy reach the maximum at L4 which drilled close to the direction of minimum horizontal stress.

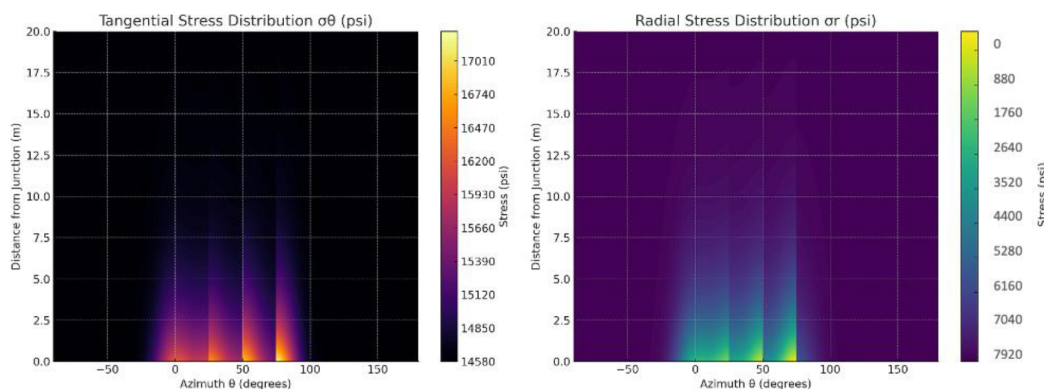


Figure 5—Tangential and radial stress distributions accounting for dynamic boundary conditions and asymmetric energy dissipation from multilateral wellbores

Tangential Stress Distribution (σ_θ in psi) Left

- Far-field stress starts at 14,600 psi and rises to ~16,500+ psi near the wellbore wall ($r = 0$).
- Each lateral introduces stress pulses.
- Due to dynamic boundary asymmetry, stress dissipation is stronger on the side opposite to previous laterals, following the "least resistance" path.
- L4 has the highest outward stress expansion, but interference is strongest between L1–L2 and L2–L3.

Right Plot: Radial Stress Distribution (σ_r in psi)

- Far-field radial stress starts at 8,000 psi and drops significantly near the wellbore.
- Attenuation and interaction are modeled oppositely to the tangential field, reflecting radial contraction effects. Energy from laterals is inwardly concentrated where interference is higher.

Areal zones controlling asymmetric pulse decay and dynamic BCs per lateral each new lateral modifies stress interference zones.

Wellbore stability analysis for multiple laterals:

Collapse and fracture gradient formulations have been modified after (Hamid and Abdalrahman 2024) adding the stress terms from equation 12.

Modified collapse gradient formulation:

$$P_{collapse,i}(r,\theta,t) = \frac{2\sigma'_h - \sigma'_v}{n} - P_0 + \frac{UCS}{n} + \beta \cdot k - \sigma_{r,total}^{(i)}(r, \theta, t) \quad (16)$$

Where: $\sigma_{r,total}^{(i)}(r, \theta, t)$ is the radial stress from equation 12 evaluated at lateral i 's spatial location, and accounts at time t , azimuth θ and distance r from the junction.

Modified fracture gradient formulation:

$$P_{fracture,i}(r,\theta,t) = (3\sigma'_h - \sigma'_v) - P_0 + T + \frac{UCS}{n} + \alpha \cdot k + \sigma_{\theta,total}^{(i)}(r, \theta, t) \quad (17)$$

Where: $\sigma_{\theta,total}^{(i)}(r, \theta, t)$ is the tangential stress from the total stress pulse superposition field (equation 12) evaluated at lateral i 's spatial location, and accounts at time t , azimuth θ and distance r from the junction. $T = \sigma_{\theta,total} - \sigma_{r,total}$ the effective shear amplification from stress pulse interactions.

Fig. 6 presents the calculated collapse and fracture pressures at each lateral (L1 to L4), using your modified equations that include:

- Stress pulse attenuation: Dynamic boundary effects using $\sigma_{r,total}^{(i)}(r, \theta, t)$ and $\sigma_{\theta,total}^{(i)}(r, \theta, t)$
- Directional distribution from the junction (based on spatial and temporal decay)
- Formation strength parameters (UCS, in-situ stresses, etc.)
- Fracture pressures (10,000 to 14,500 psi) are significantly higher than collapse pressures (5,500 to 7,000 psi).
- Collapse pressures from L1 to L4, reflecting the compounding stress interference.
- Fracture pressure increases slightly due to higher tangential stress components from cumulative dynamic effects. These results highlight how stress accumulation and redistribution from one lateral influence the wellbore stability modeling.

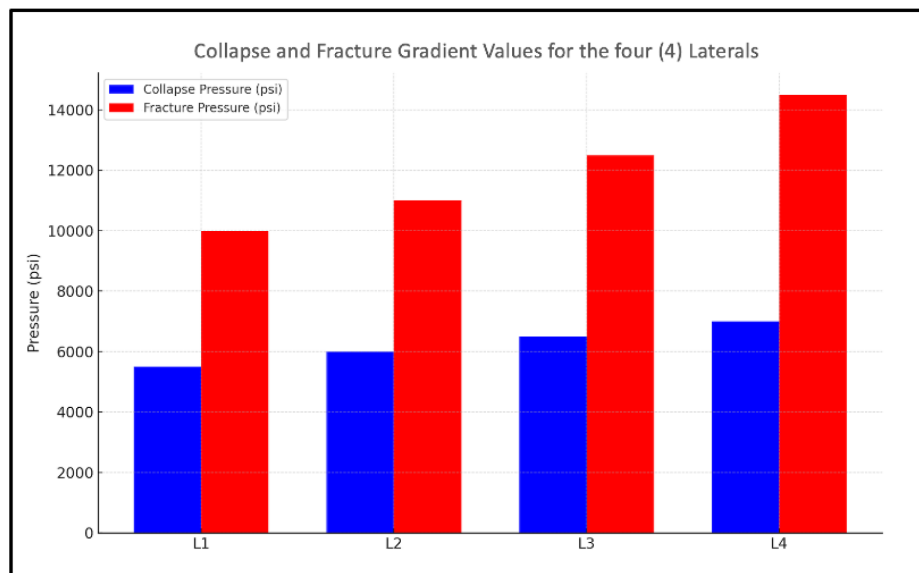


Figure 6—Collapse and fracture gradient values for 4 laterals

Fig.7 illustrates mud weight (psi) requirements plotted against true vertical depth (ft) for four multilateral wellbores (L1, L2, L3, and L4). Each curve represents the minimum mud weight required to prevent collapse; the mud weight ranges from 5500 to 7000 psi.

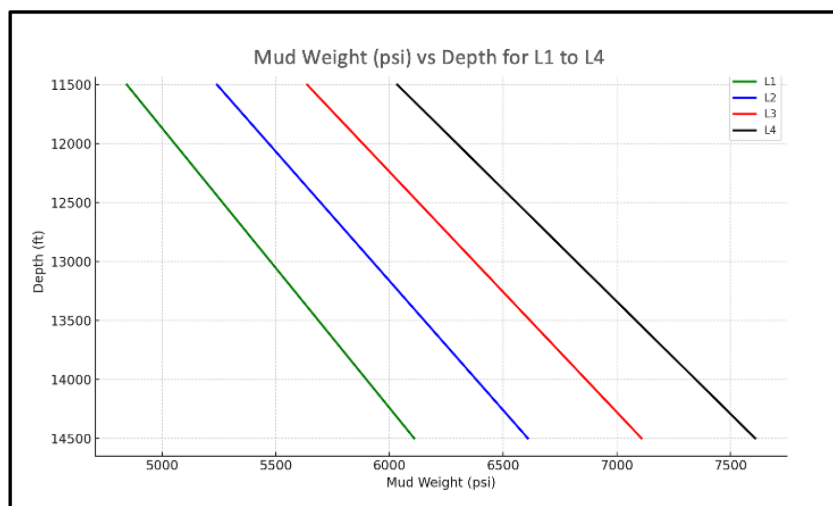


Figure 7—Recommendations of mud weights (psi) for long 4 laterals required to prevent wellbore rock failure using dynamic boundary conditions.

The figure shows, L1 requires the least mud weight at all depths, which implies that this lateral is the least affected by stress interference from adjacent laterals or boundary condition amplifications, while L4 Exhibits the highest mud weight requirement. This lateral is affected by the cumulative stress environment from all preceding laterals (L1–L3), showing maximum stress concentration due to constructive pulse superposition and dynamic boundary effects.

Fig.8 shows mud weight (pcf) requirements plotted against true vertical depth (ft) for four multilateral wellbores (L1, L2, L3, and L4), the data converted from psi to pcf.

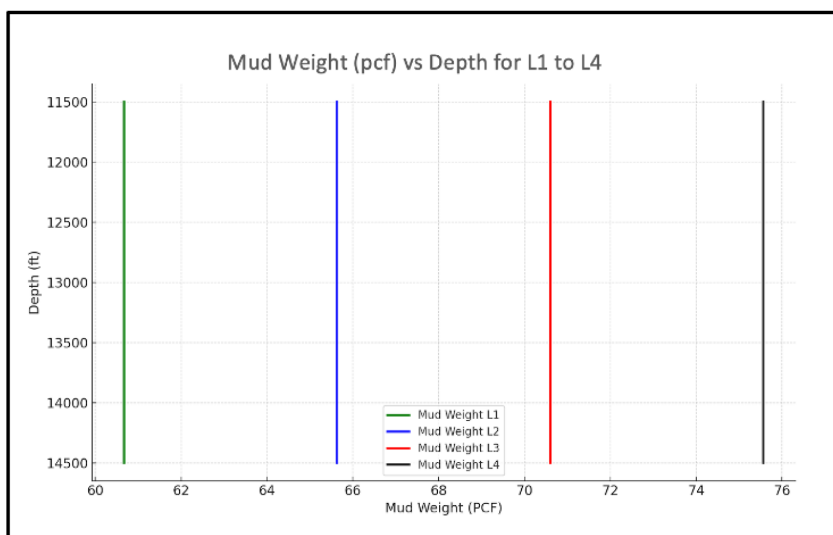


Figure 8—Recommendations of mud weights (pcf) for long 4 laterals required to prevent wellbore rock failure using dynamic boundary conditions.

Fig. 9 is the graphical representation of minimum and maximum allowable mud weights based on stress output from FEM. This provides operational guidance for safe drilling practices.

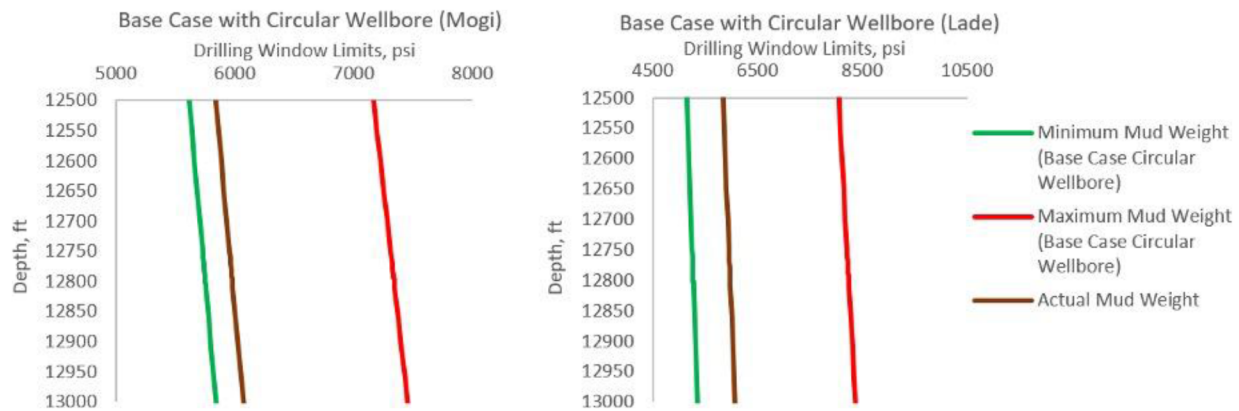


Figure 9—The recommendations of mud weights or downhole pressures required to prevent wellbore rock failure based on the output of the initial setup of the FEM (Hamid et al. 2024).

Conclusion and recommendations:

- The integrated 3D Finite Element Method (FEM) model of a multilateral wellbore system including a vertical motherbore (L0) and four lateral branches (L1–L4) spaced at 25° demonstrates how stress interference, dynamic boundary conditions (BCs), and stress pulse superposition govern wellbore stability.
- Stress Interference and Propagation: As seen in Figure 4, the overlapping stress fields visualized through stress pulse zones intensify near the junction and decrease radially outward. This attenuation is asymmetric due to preferential stress paths, highlighting the directional nature of pulse dissipation in multilateral layouts.
- Dynamic Boundary Conditions: The vertical constraint variation observed during lateral drilling alters stress distributions nonlinearly. Each new lateral introduces updated displacement and stress continuity conditions, amplifying interference zones.
- Stress Distribution Modeling: Using Equation (12), tangential and radial stress fields (Figure 5) exhibit significant spatial and azimuthal variation due to directional drilling and sequential lateral activation. The greatest stress buildup is at L4, reflecting maximum energy accumulation from previous pulses.
- Modified Stability Criteria: The incorporation of stress pulse terms in the collapse and fracture pressure gradients (Equations 16 & 17) enables localized geomechanical evaluation. Figure 6 confirm that fracture pressures are nearly twice collapse pressure, which is ranging from 10,000–14,500 psi vs. 5,500–7,000 psi.
- Mud Weight Profiles: Figures 7–9 show that L1 requires the least mud weight, whereas L4 demands the highest, a direct consequence of stress field interference from earlier drilled laterals. This establishes that cumulative interference must be considered when planning drilling sequences and mud weight management.

Recommendations:

- Strategic Lateral Sequencing:

Prioritize the drilling of laterals in the direction of maximum horizontal stress (e.g., L1) to start with minimum stress anisotropy in case of strike-slip fault regime to minimize interference amplification and collapse risk.

Angular Placement Optimization:

Maintain sufficient angular separation (e.g., 25°) between laterals to allow stress dissipation and minimize overlap of perturbation zones. Optimize azimuthal design based on in-situ stress tensors.

- Dynamic BC Integration in FEM Models:

Continuously update boundary conditions as new laterals are introduced. This allows more accurate tracking of energy redistribution and localized overstress risks.

- Customized Collapse and Fracture Models:

Use stress pulse informed gradients (as in [Equations 16](#) and [17](#)), this provides real-time risk evaluation at each lateral for collapse and fracture.

- Application to Extended Laterals and Junction Design:

Extend the model to include more complex architectures like dual junctions or stacked laterals. Use this framework to define optimum junction strength and location.

This study combines dynamic stress modeling, directional energy dissipation, and modified collapse/fracture criteria that offers a robust geomechanical strategy for multilateral drilling in complex formations. It supports field optimization, risk mitigation, and ensures structural integrity throughout the well lifecycle.

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